

Power envelopes, exact defects, and cycle restrictions for the Juggler map

Publication draft. Not submitted.

Philippe Cochin

29 August 2026

Contents

Abstract	1
1. Introduction	2
1.1 Verification convention	3
1.2 Related maps	3
2. Finite words, envelope, and defect	3
3. Inverse cells and cycles	6
4. Short descent certificates	9
5. Parity discrepancy and certified-descent densities	10
6. The remaining gap	21
7. Software archive	24
Acknowledgments	26
References	26

Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. We do not prove that conjecture.

We develop an exact finite-word calculus for the Juggler map. For every parity word w realized at n , the endpoint satisfies the power envelope

$$J^{|w|}(n)^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Hence $3^{\#O(w)} < 2^{|w|}$ forces $J^{|w|}(n) < n$ whenever $n \geq 2$. Local floor remainders lift through later exponents to an exact global defect. We prove its identity, vanishing classification, and composition law: concatenation is a two-term power-gap, and zero defect gives precisely the rigid monochrome towers. Exact inverse cells then yield necessary cycle restrictions, including superquadratic prefixes from a cycle minimum to later even states, and a small-cycle census: the Juggler map has no nontrivial cycle of length at most six. Length seven remains open.

As a secondary application, every even start and every odd start whose first image is even has a uniform one- or two-step descent certificate. A classical discrepancy estimate

$$|S_O(N)| \ll N^{5/6}, \quad S_O(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} (-1)^{\lfloor n^{3/2} \rfloor},$$

implies that this uniform short-certificate class has density $3/4$. An exact-linearization calculus for the nested floor powers extends the discrepancy method down the orbit: **every** itinerary word class of depth at most four has cardinality $2^{-|w|}N + O(N^{1-\delta_w})$ with explicit exponents. The last and hardest case, the *OOO** split, rests on a kernel theorem proved here: the exponential sum of the level-2 floor defect against smooth weights of scale $n^{9/8}$ exhibits a power saving, obtained by double Weyl differencing over an exact carry-branch decomposition. The same engine counts the contracting words of lengths five and seven, so the class of starts carrying a uniform descent certificate of length at most four, five, seven has natural density $13/16$, $7/8$, $57/64$ respectively. An unconditional counting argument shows that parity equidistribution at all depths would give the set of starts with some finite descent certificate density one; the base cases of depth at most four are now unconditional. The remaining obstacle is the level-3 kernel — the same sum one nesting deeper, where the weight scale $n^{27/16}$ exceeds n — stated here as an open problem together with a proved almost-every-shift form and a precise record of what blocks the deterministic case. None of these densities is a density of starts that reach 1.

All finite-word statements are conditional on the realized itinerary. The remaining pointwise question is whether almost every odd-to-odd start has some finite descent; no such theorem is proved here.

1. Introduction

The Juggler sequence was introduced by Pickover [1]; see also OEIS A094683 [2]. Universal arrival at 1 remains open.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \quad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities. The paper records the exact finite-word comparison, the compositional slack, the uniform short certificates, and the densities of the classes those certificates cover: all of itinerary depth four, and the contracting words of lengths five and seven.

The main contribution is the exact power-envelope and defect calculus, together with its inverse-cell and cycle consequences, chief among them a small-cycle census: no nontrivial cycle has length at most six. The certified-descent densities — $3/4$ at two steps, $13/16$ at four, $7/8$ at five, $57/64$ at seven — are secondary corollaries. We do not prove a Collatz theorem or transfer Collatz stopping-time results to J .

1.1 Verification convention

Theorems in Sections 2–4 are proved both below and in Lean under the names listed at the start of each section. The analytic estimates of Sections 5 and 6 (Theorems 5.1, 5.4, 5.7, 5.8, 5.11, 5.13–5.15, 6.4, Propositions 5.5, 6.1, Corollaries 5.12 and 5.16) are ordinary human proofs and are not formalized; the exact floor reductions beneath them — the parity bridge $\lfloor x \rfloor$ odd iff $\{x/2\} \geq 1/2$, the gap-cell identity, and the double-gap identity used by the kernel theorem — are Lean-verified (`floor_odd_iff_half_le_fract_half`, `floor_gap_eq_carry`, `seq_floor_gap`, `seq_floor_gap_second` in `GapCells.lean`). The exact-linearization lemmas of Sections 5 and 6 are additionally validated by scaled-integer computations recorded in the repository; those validations are checks, not proofs. Proposition 4.4 is an exact Python-integer census, not a theorem about an infinite set. The four-block chain named in Section 6 is Lean-certified (`four_block_pe_1999`); the failed reductions there are exact-integer counterexamples, not estimates. No finite computation is used as a termination proof.

1.2 Related maps

The nearest published comparison is the Collatz problem. Lagarias [3,4] surveys parity words and almost-all statements short of totality. Terras [5] and Everett [6] proved that almost every positive integer has a finite Collatz stopping time; Tao [7] later showed that almost all Collatz orbits attain almost bounded values. Those results are methodological cousins, not theorems about J . The Juggler analogue of Terras would be an almost-all descent theorem on the odd-to-odd class.

Crandall [8] and Matthews–Watts [9] treat piecewise-affine Hasse–Syracuse maps; the Juggler branches are floor powers, not affine. Prasad and Prasad [12] estimate excursion and stopping constants for juggler-like random walks; Section 6 keeps that comparison descriptive.

The discrepancy calculus of Section 5 concerns nested floor powers $\lfloor \lfloor n^c \rfloor^d \rfloor$. A literature check (August 2026) found no equidistribution result for such nested sequences: the Piatetski–Shapiro corpus covers single floors, intersections, and pseudo-polynomials. The tools themselves are classical — Vaaler’s trigonometric approximation to the sawtooth [13] and van der Corput’s second-derivative test in the form presented by Graham and Kolesnik [14].

2. Finite words, envelope, and defect

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w . Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters.

The identities in this section are formalized in Lean (`follows_iff_word`, `image_eq_iterate`, `image_append`, `image_monotone_of_follows`, `power_bound_word`, `power_bound_contracts`, `global_defect_identity`, `global_defect_eq_zero_iff_localsTight`, `global_defect_append`, `global_defect_eq_zero_implies_monochrome`, `power_bound_eq_iff_extremal`, `image_eq_start_defectRatio_one_plus_eta_lt_succ_sq`). The proofs below are the ordinary integer arguments.

Theorem 2.1 (fixed-word monotonicity). If $n \leq m$ and both realize w , then $J^{|w|}(n) \leq J^{|w|}(m)$.

Proof. Induct on w . The empty word is immediate. For the induction step, the two current states realize the same next letter. On the even branch, monotonicity follows from monotonicity of the integer square root; on the odd branch it follows from monotonicity of $x \mapsto x^3$ followed by the

integer square root. Apply the induction hypothesis to the prefix and then the appropriate branch inequality. $\square$

The realizing set of a fixed word need not be an interval.

Theorem 2.2 (finite-word power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Proof. Write $k = |w|$ and $o = \#O(w)$. The empty word is the equality $n \leq n$. Suppose the bound holds at a realized prefix ending at x , and the next letter is realized.

If the next letter is even, then $J(x) = \lfloor \sqrt{x} \rfloor$, so $J(x)^2 \leq x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \leq x^{2^k} \leq n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x) = \lfloor x^{3/2} \rfloor$, so $J(x)^2 \leq x^3$. Therefore

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \leq x^{3 \cdot 2^k} = (x^{2^k})^3 \leq (n^{3^o})^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. $\square$

Corollary 2.3 (exponent-gap contraction). If $n \geq 2$, w is realized at n , and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \leq n^{3^{\#O(w)}}$. The exponent gap and $n \geq 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \geq 1$, one has $m < n$. $\square$

The corollary includes familiar contracting blocks such as $OOOEE$. It does not prove that every start realizes some contracting word.

The floor slack is exact, and it is not an additive path sum. For a single branch,

$$x^e = J(x)^2 + \rho(x), \quad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \leq \rho(x) < 2J(x) + 1$. Write

$$\text{gap}(a, \rho, e) = (a + \rho)^e - a^e,$$

so $a^e + \text{gap}(a, \rho, e) = (a + \rho)^e$. If $e \geq 1$, then $\text{gap}(a, \rho, e) = 0$ if and only if $\rho = 0$, and $\rho^e \leq \text{gap}(a, \rho, e)$.

Along a realized word w , write $x_0 = n$ and $x_{j+1} = J(x_j)$, and let $\rho_j = \rho(x_j)$. Define a running slack by $D_0 = 0$ and

$$D_{j+1} = \begin{cases} D_j + \text{gap}(x_{j+1}^2, \rho_j, 2^j), & \text{next letter even,} \\ \text{gap}(x_{j+1}^2, \rho_j, 2^j) + \text{gap}(x_j^2, D_j, 3), & \text{next letter odd.} \end{cases}$$

The *global defect* is $\Delta_w(n) = D_{|w|}$. An even letter keeps the old slack and lifts the new remainder through 2^j . An odd letter first cubes the running slack and then lifts the new remainder. In particular Δ is not the sum of the local remainders.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \quad \Delta_w(n) \geq 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \geq 0$.

Proof. The empty word is $n = n + 0$. Suppose a realized prefix of length k with odd count o ends at x and satisfies $n^{3^o} = x^{2^k} + D$.

If the next letter is even, then $x = J(x)^2 + \rho(x)$, so

$$n^{3^o} = (J(x)^2 + \rho(x))^{2^k} + D = J(x)^{2^{k+1}} + D + \text{gap}(J(x)^2, \rho(x), 2^k).$$

The odd count is unchanged, and the new slack is the even update.

If the next letter is odd, then $x^3 = J(x)^2 + \rho(x)$. Cubing the inductive identity gives

$$n^{3^{o+1}} = (x^{2^k} + D)^3 = x^{3 \cdot 2^k} + \text{gap}(x^{2^k}, D, 3).$$

The leading term is

$$x^{3 \cdot 2^k} = (J(x)^2 + \rho(x))^{2^k} = J(x)^{2^{k+1}} + \text{gap}(J(x)^2, \rho(x), 2^k),$$

which is the odd update. $\square$

Theorem 2.5 (vanishing). If w is realized at n , the following are equivalent.

1. $\Delta_w(n) = 0$.
2. Every local remainder along w vanishes.
3. $J^{|w|}(n)^{2^{|w|}} = n^{3^{\#O(w)}}$.

In that case w is monochrome. More precisely, either $w = E^k$ and $n = a^{2^k}$ for an even a , or $w = O^k$ and $n = a^{2^k}$ for an odd a . A realized mixed word therefore has $\Delta_w(n) > 0$.

Proof. The identity of Theorem 2.4 gives (1) $\Leftrightarrow$ (3). For (1) $\Leftrightarrow$ (2), use the recurrence. If $e \geq 1$, a power-gap vanishes if and only if its addend vanishes. The even update is a sum of two nonnegative terms, and the odd update is a sum of two power-gaps; either vanishes only when the incoming slack and the new remainder both vanish. Thus $D_{|w|} = 0$ forces $D_0 = 0$ and every $\rho_j = 0$. The converse is immediate.

If every remainder vanishes, an even step satisfies $x_j = x_{j+1}^2$, while an odd step satisfies $x_j^3 = x_{j+1}^2$. In either case x_j and x_{j+1} have the same parity, so the itinerary is monochrome. If $w = E^k$, repeated substitution gives $n = x_k^{2^k}$; writing $a = x_k$, its parity is even. If $w = O^k$, unique factorization applied to $x_j^3 = x_{j+1}^2$ shows that every prime valuation of x_j is divisible by 2. Iterating this valuation relation shows that every prime valuation of $n = x_0$ is divisible by 2^k , hence $n = a^{2^k}$ for an odd a . Conversely, these even and odd towers make every local remainder zero. This is the extremal classification `power_bound_eq_iff_extremal`. $\square$

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then

$$\Delta_{uv}(n) = \text{gap}(m^{2^{|u|}}, \Delta_u(n), 3^{\#O(v)}) + \text{gap}(J^{|v|}(m)^{2^{|v|}}, \Delta_v(m), 2^{|u|}).$$

In particular $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$. Every local remainder satisfies $\rho_j^{2^j} \leq \Delta_w(n)$.

Proof. Write $o_u = \#O(u)$ and $o_v = \#O(v)$. Theorem 2.4 on u and on uv gives

$$n^{3^{o_u+o_v}} = (m^{2^{|u|}} + \Delta_u(n))^{3^{o_v}} = J^{|uv|}(n)^{2^{|uv|}} + \Delta_{uv}(n).$$

The first power expands as

$$m^{2^{|u|} \cdot 3^{o_v}} + \text{gap}(m^{2^{|u|}}, \Delta_u(n), 3^{o_v}).$$

Theorem 2.4 on v at m rewrites the leading term as

$$(J^{|v|}(m)^{2^{|v|}} + \Delta_v(m))^{2^{|u|}},$$

which expands as the second gap plus $J^{|uv|}(n)^{2^{|uv|}}$. Comparing the two expressions for $n^{3^{o_u+o_v}}$ yields the identity. The suffix inequality is $\rho^e \leq \text{gap}(a, \rho, e)$ on the second summand. For a local remainder at index j , split w after j letters: the suffix defect is at least ρ_j , and raising through 2^j gives the stated bound. $\square$

Composition is polynomial, not additive: later odd letters raise the prefix slack to the third power, and later letters of either parity raise a suffix slack through $2^{|u|}$. The naive recurrence $\Delta \leftarrow \Delta + \rho$ is false.

Corollary 2.7 (cycle surplus). If w is a cycle word at n , that is $J^{|w|}(n) = n$, then

$$\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$$

exactly (`image_eq_start_defectRatio`). A cycle burns its entire formal surplus in floor losses.

Proof. Theorem 2.4 with $m = n$. Nonnegativity of $\Delta_w(n)$ makes the difference a genuine one. $\square$

The identity does not supply a state-independent positive tax, and none exists: the single-step slack is provably squeezed by the scale of the state. For one realized letter at x with branch exponent $e \in \{1, 3\}$,

$$x^e < (J(x) + 1)^2$$

(`one_plus_eta_lt_succ_sq`), so the relative slack $1 + \eta = x^e / J(x)^2$ satisfies

$$\eta < \frac{2}{J(x)} + \frac{1}{J(x)^2},$$

which tends to 0 as the state grows. A uniform per-step tax is therefore impossible. The recorded extreme is an *OOE* block at $n = 180370579261640036336071806107777 \approx 1.80 \cdot 10^{32}$ whose relative slack satisfies $0 < q < 10^{-30}$; this is an exact-integer computation, not an estimate. The inequality $\Delta_w(n) > n^{3^{\#O(w)}} - n^{2^{|w|}}$ on a formally expanding word is exactly $J^{|w|}(n) < n$, so it does not forbid mixed expanding prefixes.

3. Inverse cells and cycles

The exact one-step fibers are

$$J(n) = q \iff q^2 \leq n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \leq n^3 < (m+1)^2 \quad (n \text{ odd}).$$

An odd fiber contains at most one integer (`odd_cell_unique`). Indeed, suppose $a < b$ lie in the same odd cell indexed by m . Then $(a+1)^3 \leq b^3 < (m+1)^2$ and $m^2 \leq a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

and hence $3a^2 \leq 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \leq a^3$ gives $9a^4 \leq 4m^2 \leq 4a^3$, contradicting $4a^3 < 9a^4$. An even fiber is a parity-restricted square interval and may contain many predecessors.

A nonempty realized word w with $J^{|w|}(n) = n$ is a cycle word.

The identities in this section are formalized in Lean (`cycle_word_formally_expanding`, `cycleMin_not_end_odd`, `square_scale_superquadratic`, `cycleMin_to_even_superquadratic`, `no_cycle_word_ooeoe`, `no_cycle_word_ooooee`, `no_cycle_word_length_le_six` with components `no_cycle_word_replicate_odd`, `cycleWord_rotateWord`, `cycleWord_exists_even_terminating`, `no_cycle_word_length_four_ends_even`, `no_cycle_word_length_five_ends_even`, `no_cycle_word_ooe`, `no_cycle_odd_run_append_even`, `no_cycle_word_ooeoe`). The proofs below are the ordinary integer arguments.

Theorem 3.1 (cycle restrictions). Let w be a cycle word at $n \geq 2$.

- (i) The word is formally expanding:

$$2^{|w|} < 3^{\#O(w)}$$

(`cycle_word_formally_expanding`). A contracting word cannot close a nontrivial cycle.

- (ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter (`cycleMin_not_end_odd`).

- (iii) A realized word v is *superquadratic* if

$$3^{\#O(v)} \geq 2^{|v|+1}.$$

Any realized path from a start $n \geq 2$ to a state at least n^2 is superquadratic (`square_scale_superquadratic`). On a cycle minimum the path to any later even state is superquadratic (`cycleMin_to_even_superquadratic`). The prefix *OOE* is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \leq n^{3^{\#O(w)}}$. Since $n \geq 2$, comparison of exponents gives $2^{|w|} \leq 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \geq 1$.

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start $n \geq 2$. An even state $x \geq 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \geq 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n . If the last letter were odd, the odd return cell would give $n^2 \leq x^3 < (n+1)^2$. Minimality gives $x \geq n$, hence $n^3 < (n+1)^2$, impossible for $n \geq 3$; and the minimum is odd, so $n \geq 3$.

For (iii), let a realized word v send n to $y \geq n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since $n \geq 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n . Thus $n \leq J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \leq y$, and the preceding argument applies to that prefix. $\square$

These restrictions assemble into a complete census of short cycles. Rotation reduces any cycle word to an even-terminating orientation, the expanding filter of Theorem 3.1(i) and threshold arguments eliminate every even-terminating word of length at most six except *OOOEOE* and *OOOOEE*, and those two survivors require an individual argument, which uses the last-even cell together with a coarse lower envelope. That argument is Lemma 3.2; the census is Theorem 3.3. No exclusion of cycles of length seven or more is claimed.

For $n \geq 1$,

$$n < 4 \lfloor \sqrt{n} \rfloor^2,$$

so an even step satisfies $n \leq 4J(n)^2$ and an odd step satisfies $n^3 \leq 4J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o} \leq D_v J^k(n)^{2^k}.$$

The denominator starts at 1 and updates by $D \mapsto D \cdot 4^{2^j}$ on an even letter and $D \mapsto D^3 \cdot 4^{2^j}$ on an odd letter, at step j . In particular $D_{OOO} = 2^{38}$ and $D_{OOOO} = 2^{130}$. If a cycle word ends in an even letter, the last-even cell is $J^{|w|-1}(n) < (n+1)^2$.

Lemma 3.2 (two length-six exclusions). Neither *OOOEOE* nor *OOOOEE* is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2 \cdot 256^{64}$ and $256(n+1) \leq 257n$, so $(n+1)^{64} < 2n^{64}$. Then $2^{130}(n+1)^{64} < 2^{131}n^{64}$. Since $n \geq 256 = 2^8$, one has $n^{17} \geq 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \leq n^{81}$.

For $n < 256$, neither word returns to its start. Lean checks both itinerary-and-return tables by `native_decide` on Fin 256; the same mechanism also verifies the finite numerical inequality $257^{64} < 2 \cdot 256^{64}$ used at the cutoff.

Now suppose $n \geq 256$ realizes *OOOOEE*, and write $z = J^4(n)$ for the image after the prefix *OOOO*. The last two even letters give $J(z) < (n+1)^2$ and $z < (J(z)+1)^2$, hence $z < (n+1)^4$. The lower envelope on *OOOO* is $n^{81} \leq 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $n \geq 256$ realizes *OOOEOE*. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor \sqrt{z_3} \rfloor$, so $z_3 < (y+1)^2$. The lower envelope on *OOO* is $n^{27} \leq 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters *OE* give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 257$. Then $(y+1)^3 < 2A^4$: if $y \leq A$, this follows from $(A+1)^3 < 2A^4$; if $y > A$, then $y \geq 4$ and $(y+1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \leq yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. $\square$

Theorem 3.3 (small-cycle census). No word of length at most six is a cycle word at any $n \geq 2$ (`no_cycle_word_length_le_six`). Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word (`cycleWord_rotateWord`); every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start $n \geq 2$.

If every letter is odd, the start is odd, hence $n \geq 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \geq 3$, since $x^3 \geq (x+1)^2$. The orbit ascends strictly and never returns (`no_cycle_word_replicate_odd`). Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word vE of the same length based at a cycle state $m \geq 2$ (`cycleWord_exists_even_terminating`). It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.1(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ($2 < 3^0$ and $4 < 3$ both fail). For length three the only expanding candidate is OOE : for $m \geq 5$ the next-square threshold after OO gives $J^2(m) \geq (m+1)^2$, contradicting the last-even cell $J^2(m) < (m+1)^2$; the only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd while the final even letter requires an even state. For length four the expanding filter leaves only O^3E , and for length five only O^4E ; the threshold inherits along appended odd letters, and the same last-even contradiction applies (`no_cycle_word_length_four_ends_even`, `no_cycle_word_length_five_ends_even`).

For length six the filter requires at least four odd letters among the first five, leaving O^5E , $EOOOOE$, $OEEOOE$, $OOEOOE$, $OOOEEOE$, and $OOOOEE$. The odd-run threshold excludes O^5E (`no_cycle_odd_run_append_even`). For $OOEOOE$, a minimum-based orientation meets the same next-square threshold at the internal even letter after the prefix OO , with starts below 5 checked directly (`no_cycle_word_ooeooo`). $EOOOOE$ rotates one step onto $OOOOEE$, and $OEEOOE$ rotates two steps onto $OOOEEOE$. The two remaining words are excluded by Lemma 3.2. $\square$

The census stops at length six because length seven admits even-terminating expanding words with two internal even letters that none of the recorded thresholds reach; no exclusion at length seven is claimed.

4. Short descent certificates

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$, or with image 1. Lean packages this predicate as `FiniteProgress`, an abbreviation of `DescentCertificate`. The four constructors of that type are proof forms of the same predicate, not four different claims. The predicate is existential over all finite words, not only over words of length one or two.

The theorems in this section are formalized in Lean (`even_finiteProgress`, `odd_even_finiteProgress`, `unresolved_is_odd_odd`, `reachesOne_of_all_finiteProgress`, `reachesOne_of_lt_twelve`, `even_lt_sq_twelve_reachesOne`). The proofs below are the ordinary integer arguments. Proposition 4.4 is the exact census of Section 1.1, not a Lean theorem.

Theorem 4.1 (uniform short certificates). Let $n \geq 2$.

1. If n is even, then the one-letter word E is a descent certificate: $J(n) = \lfloor \sqrt{n} \rfloor < n$.
2. If n is odd and $J(n)$ is even, then the two-letter word OE is a descent certificate.

Proof. For (1), $\lfloor \sqrt{n} \rfloor \leq \sqrt{n} < n$ for $n \geq 2$. For (2), the word OE is realized by hypothesis, and

$$J^2(n) = \lfloor \sqrt{\lfloor n^{3/2} \rfloor} \rfloor \leq n^{3/4} < n$$

for $n \geq 2$. $\square$

Theorem 4.2 (unresolved starts are odd-to-odd). If $n \geq 2$ has no descent certificate, then n is odd and $J(n)$ is odd.

Proof. Contrapositive of Theorem 4.1: an even start and an odd-to-even start each carry a short certificate. $\square$

The converse is false: many odd-to-odd starts descend after a longer word.

If every start above 1 has some descent certificate, ordinary strong induction yields arrival at 1 (`reachesOne_of_all_finiteProgress`). The hypothesis is not proved. A certificate at n only reduces the problem to a strictly smaller positive integer, which may itself be odd-to-odd.

Lean also certifies a finite landing class:

Theorem 4.3 (small residuals). Every $y \in \{1, \dots, 11\}$ reaches 1. Consequently every even residual strictly below $144 = 12^2$ reaches 1.

Proof. The orbits are finite and merge quickly: $2 \rightarrow 1$, $4 \rightarrow 2$, $6 \rightarrow 2$, $8 \rightarrow 2$, $3 \rightarrow 5 \rightarrow 11 \rightarrow 36 \rightarrow 6$, $7 \rightarrow 18 \rightarrow 4$, $9 \rightarrow 27 \rightarrow 140 \rightarrow 11$, and $10 \rightarrow 3$. Each chain ends on a start already settled, so every $y \leq 11$ reaches 1. For the second claim, an even $y < 144$ has $J(y) = \lfloor \sqrt{y} \rfloor \leq 11$, which lands in the verified set. (Lean evaluates the same finite orbits in `reachesOne_of_lt_twelve` and `even_lt_sq_twelve_reachesOne`.) $\square$

This enlarges the set of fatal landings for a hypothetical minimal counterexample.

On the complementary odd-to-odd class, first return below the start is frequent at a short horizon, but not automatic.

Proposition 4.4 (odd-to-odd first-return census). For starts $2 \leq n \leq N$ and horizon 20, write OO for the odd-to-odd starts in that range. The exact census is:

N	#OO	OO first return ≤ 20	all starts, first return ≤ 20
10^3	252	221/252 (0.877)	968/999 (0.969)
10^4	2504	2220/2504 (0.887)	9715/9999 (0.972)
10^5	24984	22379/24984 (0.896)	97394/99999 (0.974)
10^6	249926	223683/249926 (0.895)	973756/999999 (0.974)

At $N = 10^6$, 26,243 odd-to-odd starts have no return below the start in 20 steps. These rows use exact Python integer arithmetic with no size cutoff through step 20; the unresolved count is zero in every row. They are not Lean-certified and do not constitute an almost-all theorem.

5. Parity discrepancy and certified-descent densities

This section counts itinerary word classes. It proves the ambient depth-1 estimate (Theorem 5.1) with its density-3/4 corollary, then extends the method through itinerary depth four by exact linearization of the nested floor powers: depth 2 and 3 completely (Theorem 5.4, Proposition 5.5), depth 4 except the OOO^* split (Theorems 5.7 and 5.8), giving the certified-descent density $13/16$ (Corollary 5.9); then the OOO^* split itself through a kernel theorem for the level-2 floor defect (Theorems 5.11 and 5.13), completing depth 4; and finally the contracting words of lengths five and seven (Theorems 5.14 and 5.15), raising the certified density to $7/8$ and $57/64$ (Corollary 5.16).

All estimates here are human proofs; the exact floor reductions beneath them are Lean-verified (Section 1.1).

For odd n write $s(n) = (-1)^{\lfloor n^{3/2} \rfloor}$ and

$$S_O(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} s(n).$$

Let $M(N)$ be the number of odd integers $n \leq N$, and let

$$\text{OO}(N) = \#\{n \leq N : n \text{ odd}, J(n) \text{ odd}\}.$$

Then $s(n) = -1$ if and only if $J(n)$ is odd, and

$$S_O(N) = M(N) - 2 \text{OO}(N).$$

Also $M(N) = N/2 + O(1)$.

The floor sign has an exact fractional-part form. Write $x = \lfloor x \rfloor + \{x\}$. If $\lfloor x \rfloor$ is even then $\{x/2\} < 1/2$; if $\lfloor x \rfloor$ is odd then $\{x/2\} \geq 1/2$. Thus

$$\lfloor x \rfloor \text{ is odd} \iff \{x/2\} \geq \frac{1}{2}$$

(Lean: `floor_odd_iff_half_le_fract_half`). For odd $n = 2r + 1$ set

$$g(r) = \frac{(2r + 1)^{3/2}}{2}.$$

Then $s(n) = -1$ if and only if $\{g(r)\} \geq 1/2$. For R points $x_0, \dots, x_{R-1}$ and an interval $I \subset [0, 1)$, define the unnormalized interval discrepancy

$$\mathcal{D}_R(I) = \#\{0 \leq r < R : \{x_r\} \in I\} - R|I|.$$

Taking $R = M(N)$, the odd integers at most N are exactly $2r + 1$ for $0 \leq r < R$, and therefore $S_O(N) = -2\mathcal{D}_R([1/2, 1))$ for $x_r = g(r)$.

The following two lemmas are classical. The derivative estimate is the $k = 2$ case of Iwaniec–Kowalski [11, Thm. 8.20, Sec. 8.3, pp. 204–213]. The discrepancy inequality is Kuipers–Niederreiter [10, Ch. 2, Sec. 2, Thm. 2.5, pp. 112–115].

Lemma 5.A (van der Corput, second-derivative form). Let f be twice differentiable on an interval of length M , with $\lambda \leq |f''| \leq \alpha\lambda$ for some $\alpha \geq 1$. Then

$$\left| \sum e(f) \right| \ll_{\alpha} M\lambda^{1/2} + \lambda^{-1/2},$$

where the sum runs over the integers in the interval and $e(t) = e^{2\pi it}$.

Lemma 5.B (Erdős–Turán). The discrepancy of R points $x_j \in \mathbb{R}/\mathbb{Z}$ against an interval satisfies

$$D \ll \frac{R}{H} + \sum_{h=1}^H \frac{1}{h} \left| \sum_{j=1}^R e(hx_j) \right|$$

for every cutoff $H \geq 1$.

Theorem 5.1 (ambient odd-input discrepancy).

$$|S_O(N)| \ll N^{5/6}.$$

This argument is not packaged in Lean.

Proof. The second derivative is

$$g''(r) = \frac{3}{2}(2r+1)^{-1/2}.$$

It is positive and decreasing. Separate the initial term $r = 0$, then partition $1 \leq r < R$ into blocks $[M, \min(2M, R))$, where M runs over powers of 2. On such a block $g''(r) \asymp M^{-1/2}$. For the h -th Fourier mode take $f = hg$, so $\lambda \asymp hM^{-1/2}$. Lemma 5.A gives

$$\left| \sum_{r \asymp M} e(hg(r)) \right| \ll h^{1/2} M^{3/4} + h^{-1/2} M^{1/4}.$$

Lemma 5.B, with $R \asymp M$, bounds that block's contribution to $S_O(N)$ by

$$\frac{M}{H} + M^{3/4} H^{1/2} + M^{1/4}.$$

Here the middle term is $\sum_{h \leq H} h^{-1} \cdot h^{1/2} M^{3/4} \ll M^{3/4} H^{1/2}$, and the last term is $\sum_{h \leq H} h^{-1} \cdot h^{-1/2} M^{1/4} \ll M^{1/4}$. For bounded M , the trivial estimate is absorbed into the constant. Otherwise choose $H = \lfloor M^{1/6} \rfloor$, which balances the first two terms at $M^{5/6}$. The initial term contributes $O(1)$, and the geometric sum of the block bounds is $\sum_{M \leq R} O(M^{5/6}) = O(R^{5/6}) = O(N^{5/6})$. The floor sign is not replaced by a single exponential; the exponential sums are those of the sequence $\{g(r)\}$. $\square$

Corollary 5.2 (short-certificate class). The odd-to-odd starts have natural density $1/4$:

$$|\text{OO}(N) - N/4| \ll N^{5/6}.$$

Equivalently, the starts that admit the uniform one- or two-step certificates of Theorem 4.1 — every even start together with every odd-to-even start — have natural density $3/4$.

This is a density of the uniform short-certificate class of Theorem 4.1, not of all descent certificates and not of starts that reach 1.

The exact census through $N = 10^7$ has $\text{OO}(N)/N = 0.2499896$. Through $N = 10^6$ one has $S_O(N) = 146$ and running maximum 256, at $n = 985351$. A spot computation at 10^7 has running maximum 459. The observed growth is much smaller than $N^{5/6}$: the recorded maxima at 10^6 and 10^7 sit near the scale $N^{1/3}$. That empirical scale is an observation, not a theorem.

Theorem 5.1 concerns consecutive source intervals. It supplies no transfer theorem for orbit samples or sparse image sets, so it cannot by itself control the odd-to-odd dynamics after the first step. The remainder of this section does not transfer Theorem 5.1 to image sets; it attacks the nested sums directly, by removing the inner floors exactly before any analytic step.

For the rest of the section, n is odd, $X = n^{3/2}$, $m = \lfloor X \rfloor = \text{isqrt}(n^3)$, and $\theta = X - m \in [0, 1)$. At the second level, $Y = m^{3/2}$, $v = \lfloor Y \rfloor$, $\theta_2 = Y - v$; on the *OE* branch (m even), $U = m^{1/2}$, $w = \lfloor U \rfloor = J^2(n)$, and $\theta_w = U - w$. Write $\psi(x) = (-1)^{\lfloor x \rfloor}$. The first image parity of an odd start is $(-1)^m = \psi(n^{3/2})$; the next letters are the parities of v (after an odd image) or w (after an even image). Each ψ -factor below is evaluated unconditionally; the indicator algebra of the

corollaries only ever weights it on the branch where it computes the true itinerary letter, and these branch-consistency identities are machine-checked exactly on verification windows.

Lemma 5.3 (exact linearization and gap cells). (i) For every odd $n \geq 3$,

$$m^{3/2} = \frac{3}{2} m n^{3/4} - \frac{1}{2} n^{9/4} + E(n), \quad 0 \leq E(n) \leq \frac{3}{8} (n^{3/2} - 1)^{-1/2} \leq \frac{1}{2} n^{-3/4}.$$

(ii) For $h \geq 1$ and odd n , let $g(n) = m(n + 2h) - m(n)$ and $\delta(n) = (n + 2h)^{3/2} - n^{3/2}$. Then

$$g(n) = \lfloor \delta(n) \rfloor + \kappa(n), \quad \kappa(n) = [\{n^{3/2}\} \geq 1 - \{\delta(n)\}] \in \{0, 1\}.$$

Proof. (i) Let $f(t) = (X - t)^{3/2}$ on $[0, \theta]$, so $f(\theta) = m^{3/2}$. Taylor with Lagrange remainder at 0 gives $f(\theta) = X^{3/2} - \frac{3}{2} X^{1/2} \theta + \frac{3}{8} (X - \xi)^{-1/2} \theta^2$ for some $\xi \in (0, \theta)$. Substituting $\theta = X - m$ in the linear term yields $-\frac{1}{2} X^{3/2} + \frac{3}{2} m X^{1/2} = -\frac{1}{2} n^{9/4} + \frac{3}{2} m n^{3/4}$, and the remainder lies in $[0, \frac{3}{8} (X - 1)^{-1/2}]$, with $(X - 1)^{-1/2} \leq \frac{4}{3} X^{-1/2}$ already for $n \geq 3$. (ii) $g = \lfloor X + \delta \rfloor - \lfloor X \rfloor = \lfloor \delta \rfloor + \lfloor \{X\} + \{\delta\} \rfloor$, and the last floor is 1 precisely when $\{X\} + \{\delta\} \geq 1$ (Lean: `floor_gap_eq_carry`, `seq_floor_gap`). $\square$

The point of (i): the non-smooth integer m enters the phase *linearly*, multiplied by a smooth coefficient; no fractional part of growing amplitude survives. The naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2} \theta n^{3/4} + O(n^{-3/4})$ instead leaves the sawtooth θ with the growing amplitude $n^{3/4}$, which defeats the van der Corput method directly. For (ii), since δ is smooth and increasing with $\delta' \asymp h n^{-1/2}$, the level sets of $\lfloor \delta \rfloor$ partition a dyadic block $n \in (P, 2P]$ into $O(h P^{1/2})$ cells of length $\asymp P^{1/2}/h$ on which the integer part of the gap is constant.

Theorem 5.4 (nested parity discrepancy). For every $\varepsilon > 0$ and every $(a, b) \in \{0, 1\}^2 \setminus \{(0, 0)\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} \psi(n^{3/2})^a \psi(m(n)^{3/2})^b \right| \ll_{\varepsilon} N^{23/24+\varepsilon}.$$

Consequently each of the four joint parity classes of $(m, \lfloor m^{3/2} \rfloor)$ on odd $n \leq N$ has cardinality $\frac{N}{8} + O(N^{23/24+\varepsilon})$; in particular the odd-to-odd cylinder refines one letter deeper than Theorem 5.1.

Proof. Work on dyadic blocks $n \in (P, 2P]$, odd, with truncations $J_1 = J_2 = P^{1/24}$ (wave modes), $H = P^{1/12}$ (differencing), and $R = P^{1/4}$ (cell modes).

Step 1 (wave expansion). By Vaaler's theorem [13] applied to the period-2 square wave, $\psi(x) = V_J(x/2) + O(\Delta_J(x/2))$, where $V_J(t) = \sum_{0 < |q| \leq J} a_q e(qt)$ with $|a_q| \leq \min(1, 2/|q|)$ and $\Delta_J \geq 0$ is a trigonometric polynomial of degree J with constant term and coefficients $O(1/J)$. For the product, $|\psi_1 \psi_2 - V_1 V_2| \leq \Delta_1 + \Delta_2 + \Delta_1 \Delta_2$, and each error term is again a nonnegative trigonometric polynomial of the same shape. Expanding both factors reduces the theorem to bounds, uniform in the mode pair, for

$$S_{\mu, \nu}(P) = \sum_{n \in (P, 2P], n \text{ odd}} e(\mu n^{3/2} + \nu m^{3/2}), \quad |\mu| \leq J_1, \quad \frac{1}{2} \leq |\nu| \leq J_2,$$

with $\mu, \nu \in \frac{1}{2} \mathbb{Z}$; the pure first-factor majorant sums are $\ll P/J_1 + P^{5/6+\varepsilon} \ll P^{23/24+\varepsilon}$ by the depth-1 machinery, and mode weights sum to $O(\log^2 P)$. Write $i = 2\mu$, $j = 2\nu$.

Step 2 (linearization). By Lemma 5.3(i), replacing $\frac{j}{2} m^{3/2}$ by $\frac{3j}{4} m n^{3/4} - \frac{j}{4} n^{9/4}$ changes $S_{\mu, \nu}$ by at most $2\pi \frac{j}{4} \sum_{n \sim P} n^{-3/4} \ll j P^{1/4} \leq P^{7/24}$.

Step 3 (van der Corput A-process). With $H = P^{1/12}$, $|S_{\mu, \nu}|^2 \leq \frac{2P}{H} \sum_{0 \leq h < H} |T_h|$, where T_h sums $e(\Phi(n+2h) - \Phi(n))$ over the overlap and Φ is the linearized phase. The exact rewrite

$m(n+2h)(n+2h)^{3/4} - m(n)n^{3/4} = g(n)(n+2h)^{3/4} + m(n)[(n+2h)^{3/4} - n^{3/4}]$ with $m(n) = n^{3/2} - \theta_n$ gives

$$\Phi(n+2h) - \Phi(n) = A_h(n) + \frac{3j}{4} g(n) (n+2h)^{3/4} + O(jhP^{-1/4}),$$

where A_h is smooth and the error is the $-\frac{3j}{4}\theta_n[(n+2h)^{3/4} - n^{3/4}]$ term. The two $n^{5/4}$ -scale contributions of A_h cancel at leading order, leaving $A_h'' = \frac{81}{256}jh^2n^{-7/4}(1+o(1))$ plus an i -part $O(ihP^{-3/2})$, both $o(jhP^{-3/4})$ in these ranges.

Step 4 (cells). Split T_h over the $O(hP^{1/2})$ cells of Lemma 5.3(ii). On a cell, $g = G + \kappa$ with G constant and $\kappa(n) = [\{n^{3/2}\} \geq \rho(n)]$, $\rho = 1 - \{\delta(n)\}$ smooth and monotone there. Vaaler-expand κ with modes $e(rn^{3/2})$, $0 < |r| \leq R$. The moving endpoint costs nothing: since $\rho(n) = 1 + G - \delta(n)$, the coefficient's n -dependent piece contributes the exact smooth phase $r\delta(n)$, and $e(rn^{3/2} + r\delta(n)) = e(r(n+2h)^{3/2})$ up to a unimodular constant. Every r -term therefore falls into one of the two smooth families $e(rn^{3/2})$, $e(r(n+2h)^{3/2})$; the main term $\delta(n) - G$ has total variation exactly 1 per cell and is absorbed by one Abel summation.

Step 5 (second-derivative test per cell). The remaining phases are smooth. For the $r = 0$ parts, $|f''| \asymp \lambda_0 = jhP^{-3/4}$ with a single sign and bounded ratio across a cell; van der Corput II [14] on a cell of length $\ell \asymp P^{1/2}/h$ gives $\ll \ell\lambda_0^{1/2} + \lambda_0^{-1/2}$, hence over all cells $\ll (jh)^{1/2}P^{5/8} + j^{-1/2}h^{1/2}P^{7/8}$. For $r \neq 0$, the curvature is dominated by the r -term $\asymp |r|P^{-1/2}$ of fixed sign (as $jh \leq P^{1/8} \ll P^{1/4}$); the same test and the $1/|r|$ weights give $\ll R^{1/2}P^{3/4} + hP^{3/4}\log P$, and the majorant errors contribute $O(P/R) = O(P^{3/4})$. In total $|T_h| \ll P^{7/8}(1 + h^{1/2})$, uniformly in the modes.

Step 6 (assembly). $|S_{\mu,\nu}|^2 \ll P^2/H + P^{15/8}H^{1/2} \ll P^{23/12}$ at $H = P^{1/12}$, so $|S_{\mu,\nu}| \ll P^{23/24}$. Summing mode weights, majorants, and dyadic blocks gives the theorem with $N^{23/24}\log^3 N$. $\square$

The exponent 23/24 is deliberately unoptimized; every stage has slack. The observed cancellation is far stronger (envelope exponent ≈ 0.28 in exact censuses), which the proof does not claim.

Proposition 5.5 (OE-branch third letter). For $a \in \{0, 1\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} ((-1)^m)^a \psi(m^{1/2}) \right| \ll_\varepsilon N^{7/8+\varepsilon},$$

and consequently $\#\text{OEO}(N)$, $\#\text{OEE}(N) = N/8 + O(N^{7/8+\varepsilon})$. Together with Theorem 5.4 this makes depth 3 complete: each of OOO , OOE , OEO , OEE has density $1/8$ with a power saving.

Proof. Taylor expansion of $(X - \theta)^{1/2}$ with both correction terms one-signed gives

$$m^{1/2} = n^{3/4} + D_1(n), \quad -\frac{1}{2}X^{-1/2} - \frac{1}{8}(X-1)^{-3/2} \leq D_1 \leq 0.$$

D_1 is decaying, so replacing $\frac{l}{2}m^{1/2}$ by $\frac{l}{2}n^{3/4}$ inside a Vaaler mode costs $\ll l \sum_{n \sim P} n^{-3/4} \ll lP^{1/4}$, absorbable at $l \leq 2P^{1/24}$. The mode sums are then pure: $\sum_{n \sim P} e(\frac{i}{2}n^{3/2} + \frac{l}{2}n^{3/4})$ with $l \neq 0$. For $i \neq 0$, $\varphi'' \asymp |i|P^{-1/2}$ is single-signed and van der Corput II gives $\ll i^{1/2}P^{3/4} + i^{-1/2}P^{1/4}$; for $i = 0$, $\varphi'' \asymp lP^{-5/4}$ gives $\ll l^{1/2}P^{3/8} + l^{-1/2}P^{5/8}$. Vaaler majorants, truncation tails, and dyadic assembly are as in Theorem 5.4, and every bound is $\ll P^{7/8}$ after summing mode weights. Branch consistency: the $(1 + (-1)^m)/2$ factor vanishes exactly on odd m , where J^2 takes the $3/2$ -power branch, so the unconditional $\psi(m^{1/2})$ is only weighted where it computes the true third letter. $\square$

Lemma 5.6 (fourth-letter linearization). For every odd $n \geq 3$,

$$v^{1/2} = n^{9/8} + D(n), \quad -\frac{3}{4}n^{-3/8} - n^{-9/8} \leq D(n) \leq 0.$$

Proof. Two applications of the Lemma 5.3(i) pattern. With $f(t) = (Y - t)^{1/2}$ on $[0, \theta_2]$: $v^{1/2} = m^{3/4} - \frac{1}{2}\theta_2 m^{-3/4} - E_2$, $0 \leq E_2 \leq \frac{1}{8}(Y - 1)^{-3/2}$. With $f(t) = (X - t)^{3/4}$ on $[0, \theta]$: $m^{3/4} = n^{9/8} - \frac{3}{4}\theta n^{-3/8} - E_3$, $0 \leq E_3 \leq \frac{3}{32}(X - 1)^{-5/4}$. Every term is nonpositive after the leading one, and $\frac{1}{2}m^{-3/4} + E_3 + E_2 \leq n^{-9/8}$ for $n \geq 3$. $\square$

Unlike the depth-2 layer, *every* non-smooth term now has decaying amplitude: the cumulative phase cost of replacing $\frac{k}{2}v^{1/2}$ by the smooth $\frac{k}{2}n^{9/8}$ over a dyadic block is $\ll kP^{5/8}$, negligible against the target.

Theorem 5.7 (triple parity discrepancy). For every $\varepsilon > 0$ and every $(a, b, c) \in \{0, 1\}^3 \setminus \{(0, 0, 0)\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} \psi(n^{3/2})^a \psi(m^{3/2})^b \psi(v^{1/2})^c \right| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the eight sign classes of $(\psi(n^{3/2}), \psi(m^{3/2}), \psi(v^{1/2}))$ on odd $n \leq N$ has cardinality $\frac{N}{16} + O(N^{23/24+\varepsilon})$.

Proof. The $c = 0$ cases are Theorem 5.4 and Theorem 5.1. For $c = 1$, wave-expand all present sign factors as in Step 1 of Theorem 5.4, with truncations $J_1 = J_2 = J_3 = P^{1/24}$, and bound

$$S_{i,j,k}(P) = \sum_{n \in (P, 2P], n \text{ odd}} e\left(\frac{i}{2}n^{3/2} + \frac{j}{2}m^{3/2} + \frac{k}{2}v^{1/2}\right)$$

uniformly for $|i| \leq 2J_1$, $|j| \leq 2J_2$, $1 \leq |k| \leq 2J_3$. By Lemma 5.6, replacing $\frac{k}{2}v^{1/2}$ by $\frac{k}{2}n^{9/8}$ costs $\ll |k|P^{5/8} \leq P^{2/3}$, before any differencing.

If $j = 0$, the remaining sum is a single smooth exponential sum with phase $\frac{i}{2}n^{3/2} + \frac{k}{2}n^{9/8}$. Both curvature terms are positive for positive modes, and for mixed signs with $|i| \geq 1$ the first dominates by $P^{3/8-1/24}$; van der Corput II gives $\ll P^{5/8}$ for $i = 0$ and $\ll |i|^{1/2}P^{3/4} + P^{1/4}$ otherwise, both $\ll P^{23/24}$.

If $j \neq 0$, the phase is exactly the Theorem 5.4 phase plus the smooth passenger $\frac{k}{2}n^{9/8}$. Steps 2–6 run verbatim; the passenger modifies the smooth part of the differenced phase by $\frac{k}{2}[(n+2h)^{9/8} - n^{9/8}]$, whose second derivative $\ll |k|hP^{-15/8}$ is smaller than the retained cell curvature $jhP^{-3/4}$ and the r -mode curvature $|r|P^{-1/2}$ by powers of P . Every sign-dominance check of Theorem 5.4 holds with these margins. $\square$

Theorem 5.8 (the OE splits).**

$$\#\text{OEOE}(N), \#\text{OEEO}(N) = \frac{N}{16} + O(N^{7/8+\varepsilon}), \quad \#\text{OEEE}(N), \#\text{OEEO}(N) = \frac{N}{16} + O(N^{13/16+\varepsilon}).$$

Together with Theorems 5.4 and 5.7 this proves every depth-4 itinerary word class except $OOO*$.

Proof. On the OE branch m is even and $w = \lfloor U \rfloor$, $U = m^{1/2}$. The w -level linearization is Lemma 5.3(i) verbatim with base U : since $Um^{1/4} = m^{3/4}$ exactly,

$$w^{3/2} = m^{3/4} - \frac{3}{2}m^{1/4}\theta_w + E, \quad 0 \leq E \leq \frac{3}{8}(U - 1)^{-1/2},$$

and $m^{3/4} = n^{9/8}$ up to one-signed corrections of scale $n^{-3/8}$. For a Vaaler mode $k \neq 0$, the phase is therefore $\varphi_1(n) - B(n)\theta_w(n)$ up to absorbable errors, with $\varphi_1 = \frac{i}{2}n^{3/2} + \frac{j}{2}n^{3/4} + \frac{k}{2}n^{9/8}$ and $B = \frac{3k}{4}n^{3/8}(1 + O(n^{-3/2}))$: a *single growing sawtooth* of amplitude $\asymp kn^{3/8}$ riding $\theta_w = \{m^{1/2}\}$, whose underlying integer increments once every $\asymp \frac{4}{3}n^{1/4}$ steps. Truncations: $J_1 = J_2 = J_3 = P^{1/8}$.

Drift-1 intervals. $B' \asymp kn^{-5/8}$, so B drifts by at most 1 on intervals I of length $L_0 = P^{5/8}/k$; there are $\asymp kP^{3/8}$ of them. On I , expand $e(-B\{U\}) = \sum_r a_r(B)e(rU)$ (Fourier in U , Vaaler truncation $|r+B| \leq T = P^{5/16}$); the coefficients satisfy $|a_r(B)| \leq \min(1, |r+B|^{-1})$ with window mass $O(\log T)$, and their n -dependence through $B(n)$ has total variation $O(1)$ per interval, removed by one partial summation per mode.

Mode sums. Smoothing $rU \rightarrow rn^{3/4}$ costs $kP^{5/8}$ in total. Writing $r = -B_0 + t$, $|t| \leq T$, the curvature of the mode phase is

$$\varphi'' = \frac{27k}{128}n^{-7/8} - \frac{3}{16}(t + \frac{j}{2})n^{-5/4} + \frac{3i}{8}n^{-1/2}.$$

For $i \neq 0$ the last term dominates and van der Corput II gives $\ll i^{1/2}P^{3/4} + i^{-1/2}kP^{5/8}$ after summing intervals. For $i = 0$, $(T + J_2)P^{-5/4} \ll kP^{-7/8}$ because $T = P^{5/16} \ll kP^{3/8}$ for every $k \geq 1$, so $\lambda \asymp kn^{-7/8}$ is single-signed; van der Corput II per interval and summation over $\asymp kP^{3/8}$ intervals give $S_k \ll k^{1/2}P^{13/16+\varepsilon}$. With Vaaler weights $1/k$, $\sum_{k \leq J_3} k^{-1/2}P^{13/16} = J_3^{1/2}P^{13/16} = P^{7/8}$, balanced against the truncation error $P/J_3 = P^{7/8}$ at $J_3 = P^{1/8}$. Total $\ll P^{7/8+\varepsilon}$.

OEE branch. The fourth letter needs $\psi(w^{1/2})$, and $w^{1/2} = U^{1/2} - \frac{1}{2}\theta_w U^{-1/2} - \frac{1}{8}\theta_w^2(U - \xi)^{-3/2}$ has *decaying* amplitudes only, with $U^{1/2} = m^{1/4} \rightarrow n^{3/8}$ likewise. The pure modes $e(\frac{l}{2}n^{3/8})$ have $\lambda \asymp lP^{-13/8}$, giving $l^{1/2}P^{3/16} + l^{-1/2}P^{13/16}$; the mixed cases sit in the dominance hierarchy $iP^{-1/2} \gg jP^{-5/4} \gg lP^{-13/8}$ with all three second derivatives of the same sign. Total $\ll N^{13/16+\varepsilon}$. $\square$

Corollary 5.9 (depth-4 census and certified-descent density). Every itinerary word class of depth at most four except OOO^* satisfies $\#\{n \leq N : \text{word}(n) \text{ has prefix } w\} = 2^{-|w|}N + O(N^{1-\delta_w})$ with the explicit exponents above. In particular

$$\#\{n \leq N : n \text{ odd, itinerary } OOOE\} = \frac{N}{16} + O(N^{23/24+\varepsilon}),$$

and the class of starts with a certified descent within four steps — evens (one step), OE (two steps), $OOEE$ (four steps) — has cardinality

$$\frac{N}{2} + \frac{N}{4} + \frac{N}{16} + O(N^{23/24+\varepsilon}) = \frac{13N}{16} + O(N^{23/24+\varepsilon}).$$

Proof. For $OOEE$, the indicator algebra gives $\#OOEE = \frac{1}{8} \sum_{n \leq N \text{ odd}} (1 - \psi_1)(1 + \psi_2)(1 + \psi_3)$ with $\psi_1 = \psi(n^{3/2})$, $\psi_2 = \psi(m^{3/2})$, $\psi_3 = \psi(v^{1/2})$; expanding gives the main term and seven sign sums bounded by Theorem 5.7. The factor $(1 + \psi_2)$ vanishes precisely where J^3 would take the odd branch, so the unconditional ψ_3 is only weighted where it computes the true fourth letter. Every $OOEE$ start descends within four steps by the power envelope: $3^2 < 2^4$, so Corollary 2.3 applies. The even class and the OE class carry the certificates of Theorem 4.1, and the three classes are disjoint. The remaining depth- ≤ 4 prefixes are Theorem 5.1 (depth 1), Theorem 5.4 and Proposition 5.5 (depths 2–3), and Theorems 5.7–5.8 (depth 4). $\square$

The density $13/16$ is the exact ceiling of this one-growing-layer machinery: a word contracts iff $3^o < 2^\ell$, the method so far proves letters at positions 1–3 of any word plus further letters along even branches only, and the contracting minimal words with all odd letters at positions ≤ 2 are exactly E , OE , and $OOEE$, of total density $\frac{1}{2} + \frac{1}{4} + \frac{1}{16} = \frac{13}{16}$. Any further certified density requires the OOO^* split — a second growing layer, where the fourth-letter phase coefficient $W \asymp kn^{9/8}$ crosses integers within single steps and no drift-1 interval exists. The rest of this section closes that split and harvests the contracting words it unlocks.

Every reorganization of the OOO^* phase funnels into one object: for smooth weights c of scale $kP^{9/8}$ on $n \sim P$,

$$K_c(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n) \{ \lfloor n^{3/2} \rfloor^{3/2} \}).$$

The next lemma collects the exact reductions behind its treatment: the kernel phase is the level-2 local floor defect; the second difference of the level-2 integer obeys an exact floor identity; and on an explicit carry-branch decomposition that second difference is a smooth function with a frozen floor.

Lemma 5.10 (level-2 defect, double gap, and branch freeze). Fix shifts $d_1 = 2h_1$, $d_2 = 2h_2$ and write Δ_1 , Δ_2 , $\Delta\Delta$ for the corresponding difference operators in n , and $W = \Delta_1 Y$.

(i) For odd $n \geq 3$,

$$\frac{3}{4} v^{1/2} \theta_2 = \frac{1}{2} (m^{9/4} - v^{3/2}) - R, \quad 0 \leq R \leq \frac{3}{16} v^{-1/2}.$$

Hence for $c = \frac{3k}{4} v^{1/2}$ the kernel phase $c \theta_2$ equals $\frac{k}{2} (Y^{3/2} - v^{3/2})$ up to $kR \ll kP^{-9/8}$ per term: the kernel is the exponential sum of the level-2 local floor defect.

(ii) Exactly, with $g_2 = \Delta_1 v$,

$$\Delta_2 g_2 = \lfloor \Delta \Delta Y \rfloor + \kappa'' + \Delta_2 \kappa_2, \quad \kappa_2 = [\theta_2 \geq 1 - \{W\}], \quad \kappa'' = [\{W\} \geq 1 - \{\Delta \Delta Y\}],$$

and every carry is a difference of unit sawtooths: $[\{A\} + \{B\} \geq 1] = \{A\} + \{B\} - \{A+B\}$ (Lean: `seq_floor_gap_second`).

(iii) Write $b_1 = \lfloor \Delta_1 X \rfloor$, $b_2 = \lfloor \Delta_2 X \rfloor$, $b_{12} = \lfloor \Delta_{12} X \rfloor$ (shift $d_1 + d_2$) — each constant on runs of length $\asymp P^{1/2}/h$ — and let $\kappa = (\kappa_1, \kappa_2, \kappa_{12}) \in \{0, 1\}^3$ be the level-1 carries, so that the shifted values of m are $m + b_1 + \kappa_1$, $m + b_2 + \kappa_2$, $m + b_{12} + \kappa_{12}$. On each b -run intersection and carry branch,

$$\Delta \Delta Y = F_\kappa(m), \quad F_\kappa(m) = (m + b_{12} + \kappa_{12})^{3/2} - (m + b_1 + \kappa_1)^{3/2} - (m + b_2 + \kappa_2)^{3/2} + m^{3/2}$$

exactly. The net offset $j = (b_{12} + \kappa_{12}) - (b_1 + \kappa_1) - (b_2 + \kappa_2)$ satisfies $|j| \leq 3$ for $h_1 h_2 \leq P^{1/2}/3$, and

$$F_\kappa \asymp \frac{3}{2} |j| P^{3/4} + h_1 h_2 P^{1/4}, \quad F'_\kappa(m) \asymp |j| P^{-3/4} + h_1 h_2 P^{-5/4} < 1,$$

so $\lfloor F_\kappa(X(n)) \rfloor$ is constant on runs of length $\asymp \min(P^{1/4}/|j|, P^{3/4}/(h_1 h_2))$. The branch indicator is a finite union of arcs in the single variable θ with slowly moving endpoints (drifts $\asymp hP^{-1/2}$): exactly the moving-endpoint pattern of Step 4 of Theorem 5.4.

Proof. (i) Taylor of $(v + \theta_2)^{3/2}$ at v , with $Y^{3/2} = m^{9/4}$. (ii) The gap identity of Lemma 5.3(ii) applied twice — to Y at shift d_1 , then to the real sequence $n \mapsto W(n)$ at shift d_2 , with $\Delta_2 W = \Delta \Delta Y$; the sawtooth form of the carry is the Lean-verified floor-carry identity rearranged. (iii) The corner identities are Lemma 5.3(ii) at the three shifts. The offset bound follows from $|\Delta \Delta X| \leq 4h_1 h_2 \sup |X''| = 3h_1 h_2 P^{-1/2}$. The derivative bound is the mean value theorem on the second difference of $\frac{3}{2} m^{1/2}$ plus $\frac{3}{4} j(m + \xi)^{-1/2}$. $\square$

One warning, recorded as negative knowledge: $\lfloor \Delta \Delta Y \rfloor$ itself is *not* frozen. The level-1 carry toggles at essentially every step and shifts $\Delta \Delta Y$ by $\frac{3}{2} j_1 m^{1/2}$; in exact windows its mean run length is 1.5 with jumps of size $\frac{3}{2} P^{3/4}$. The branch organization of (iii), which carries the flicker inside the indicator, is forced.

Theorem 5.11 (kernel cancellation). Let c be smooth on $(P, 2P]$ with $c^{(r)} \asymp kP^{9/8-r}$ for $r = 0, \dots, 4$, derivative signs following the monomial pattern (e.g. $c = \frac{3k}{4}n^{9/8}$). Then

$$K_c(P) \ll P^{1-\delta+\varepsilon}, \quad \delta = \frac{1}{64} \text{ for } k \ll P^\varepsilon, \quad \delta = \frac{1}{72} \text{ uniformly for } k \leq P^{1/24}.$$

Proof. Work on a dyadic block $n \sim P$, odd.

Step 1 (double Weyl differencing). With $H_1 = P^a$, $H_2 = P^b$ chosen in Step 6,

$$|K_c|^2 \ll \frac{P^2}{H_1} + \frac{P}{H_1} \sum_{h_1 \leq H_1} |T_1(h_1)|, \quad |T_1|^2 \ll \frac{P^2}{H_2} + \frac{P}{H_2} \sum_{h_2 \leq H_2} |T_2(h_1, h_2)|,$$

where $T_2 = \sum_n e(\varphi_2)$ and $\varphi_2 = \Delta\Delta(c\theta_2) = \Delta\Delta(cY) - \Delta\Delta(cv)$: an exact split into a real-analytic block and an integer block.

Step 2 (exact product rule). For both blocks,

$$\Delta\Delta(cf) = c_{11} \Delta\Delta f + (\Delta_2 c)(n+d_1) \Delta_1 f + (\Delta_1 c)(n+d_2) \Delta_2 f + (\Delta\Delta c) f,$$

with $\Delta_i c \asymp kh_i P^{1/8}$ and $\Delta\Delta c \asymp kh_1 h_2 P^{-7/8}$.

Step 3 (the Y-block). Split into the $O((h_1+h_2)P^{1/2})$ cell intersections of Lemma 5.3(ii); within a cell every shifted Y -value is a smooth function of the single variable $m = X - \theta$, so no $\Delta\theta$ cross-terms arise. The $(\Delta\Delta c)Y$ and $(\Delta_i c)\Delta_j Y$ pieces have smooth curvature $\asymp kh_1 h_2 P^{-5/8}$ and θ -coefficients $\asymp kh_1 h_2 P^{-1/8} < 1$ under the constraint (C1) $kh_1 h_2 \leq P^{1/8}$; sub-unit sawtooths are expanded in Fourier modes of θ at multiplicative mode-mass $O(\log P)$. The $c_{11}\Delta\Delta Y$ piece splits over the branches of Lemma 5.10(iii): per branch the smooth part $cF_\kappa(X)$ has curvature $\asymp k|j|P^{-1/8} + kh_1 h_2 P^{-5/8} < 1$, and the θ -content $cF'\theta$ has coefficient $\asymp k|j|P^{3/8}$ with window drift $(cF')' \asymp k|j|P^{-5/8} < 1$; a shifted-window expansion produces X -modes of size $s \asymp k|j|P^{3/8}$ with curvature $sX'' \asymp k|j|P^{-1/8} < 1$, handled by van der Corput II.

Step 4 (the v-block). v is an integer, so no fractional part of c is ever split off. The $(\Delta\Delta c)v$ piece reduces to Step 3 plus a θ_2 -coefficient $\asymp kh_1 h_2 P^{-7/8} < 1$. The $(\Delta_2 c)\Delta_1 v$ piece is $\Delta_2 c(W - \{W\})$ plus a carry: $\Delta_2 c \cdot W$ is smooth per cell; the sawtooth $-\Delta_2 c\{W\}$ has window drift $(\Delta_2 c)' \asymp kh_2 P^{-7/8} < 1$ and expands into W -modes $r \lesssim kh_2 P^{1/8}$, each again smooth per cell; the carry weight $\kappa_2 = \{Y\} + \{W\} - \{Y+W\}$ (Lemma 5.10(ii)) expands into unit sawtooths of Y - and W -forms with Vaaler windows $|r| \leq R = P^\rho$ at majorant cost P/R per layer; the Y -modes linearize (Lemma 5.3(i) pattern) to X -modes $s \asymp RP^{3/4}$ of curvature $sX'' \asymp RP^{1/4} \gg 1$ — pieces without a frozen-floor factor take the third-derivative test over full ranges ($\lambda_3 = sX''' \asymp RP^{-3/4} < 1$, saving $P^{1/8}/R^{1/6}$); pieces riding a frozen floor are the mixed class of Step 5. The main piece $c_{11}\Delta\Delta v = c_{11}\Delta_2 g_2$ is Lemma 5.10(ii): per branch the frozen integer $J_F = \lfloor F_\kappa(X(n)) \rfloor$ combines with the $c_{11}\Delta\Delta Y$ -content into the smooth per-run phase $c(F_\kappa - J_F)$, of curvature $\asymp k|j|P^{-1/8} + kh_1 h_2 P^{-5/8}$, single-signed per branch; the carries κ'' and $\Delta_2 \kappa_2$ are indicator weights times $e(\mp c)$.

Step 5 (dominance and mode assembly). The final pieces fall into four classes. (i) *Pure smooth pieces*: van der Corput II per frozen run; the leading curvature per branch is a double difference of one smooth function, so the double mean value theorem gives a single sign — for the monomial family the composite check is $\alpha(\alpha-1)(\alpha+\beta-2)(\alpha+\beta-3) > 0$ at $\alpha = \frac{9}{8}$, $\beta \in \{\frac{1}{4}, \frac{3}{4}\}$, with composite exponents $\frac{15}{8}, \frac{11}{8} \notin \{0, 1, 2\}$. Per-run application is legitimate because $\lambda_2^{-1/2} \asymp P^{1/16} \ll P^{1/4} \asymp$ run length; saving $\geq P^{1/16}/(k|j|)^{1/2}$. (ii) *Slow-mode pieces*: van der Corput II over full ranges; dominance is clean because the colliding curvature scale $s^* \asymp kP^{3/8}$ lies outside every mode window.

(iii) *Mixed pieces* — a frozen-floor factor times a large X -mode $s \asymp rP^{3/4}$: van der Corput II fails there ($\lambda_2 \gg 1$) and the per-run third-derivative test returns only the trivial bound. The repair is one *targeted third Weyl differencing* (shift $2h_3$, $h_3 \leq H_3$) followed by the split $J_F = F - \{F\}$: the differenced coefficient $\Delta_3 c \asymp kh_3 P^{1/8}$ has window drift < 1 against the *slow* sawtooth $\{F(X)\}$ (drift $\asymp P^{-1/4}$), so the shifted-window expansion applies with no run segmentation; $\Delta_3[F]$ is bounded with slow indicators; and the differenced mode curvature $s(\Delta_3 X)'' \asymp rh_3 P^{-3/4} < 1$ is single-signed and dominant. Van der Corput II over full ranges and the balance $H_3 = r^{-1/3} P^{1/4}$ recover the class-(iv) saving. (iv) *Carry-mode pieces without a frozen floor*: the third-derivative test over full ranges, $\lambda_3 \asymp rP^{-3/4}$, saving $P^{1/8}/r^{1/6}$.

Step 6 (assembly). Sub-unit sawtooths cost multiplicative mode-mass $O(\log P)$ only; mode masses multiply over at most four expansion layers plus the targeted differencing: P^ε . With $R = P^{1/16}$, the savings $P^{1/16}/(k|j|)^{1/2}$ and $P^{1/8-1/96}$ and the truncation $P^{-1/16}$ balance at $|T_2| \ll P^{1-1/16+\varepsilon}$ under (C1) and $h_1 h_2 \leq P^{1/2}/3$. Unwinding the two differencings: with $k \leq P^{\kappa_0}$ and $a+b \leq \frac{1}{8} - \kappa_0$, $|K_c| \ll P(P^{-a/2} + P^{-b/4} + P^{-1/64})$. For $\kappa_0 = \varepsilon$: $a = \frac{1}{32}$, $b = \frac{1}{16}$, $\delta = \frac{1}{64}$. For $\kappa_0 = \frac{1}{24}$: $a = \frac{1}{36}$, $b = \frac{1}{18}$, $\delta = \frac{1}{72}$. Exponents deliberately unoptimized. $\square$

The proof was re-derived adversarially step by step; exact-phase probes at the first, second, and third differenced levels show square-root cancellation throughout (repository), far stronger than the bound claims.

Corollary 5.12 (low-exponent families). The bound of Theorem 5.11 holds for every monomial-pattern family $c^{(r)} \asymp kP^{\alpha-r}$ with $0 < \alpha \leq \frac{9}{8}$, uniformly for $k \leq P^{1/24}$.

Proof. Every constraint in the proof is monotone in α on $(0, \frac{9}{8}]$: (C1) relaxes to $kh_1 h_2 \leq P^{5/4-\alpha}$; every curvature and window drift shrinks; the sign-dominance product $\alpha(\alpha-1)(\alpha+\beta-2)(\alpha+\beta-3) > 0$ holds at $\beta \in \{\frac{1}{4}, \frac{3}{4}\}$ throughout the interval; the assembly exponents only improve. $\square$

Theorem 5.13 (the OOO* splits; depth four complete). For $w \in \{OOOE, OOOO\}$,

$$\#\{n \leq N \text{ odd} : \text{word}_4(n) = w\} = \frac{N}{16} + O(N^{1-1/72+\varepsilon}).$$

Hence **every** itinerary word class of depth at most four satisfies $\#\{n \leq N\} = 2^{-|w|}N + O(N^{1-\delta_w})$ with explicit $\delta_w > 0$: depth-4 parity equidistribution is complete.

Proof. The class indicator expands (Vaaler, truncation $J_3 = P^{1/24}$, error P/J_3) into mode sums $S_{ijk} = \sum_n e(\frac{i}{2}X + \frac{j}{2}Y + \frac{k}{2}v^{3/2})$ over dyadic blocks. Three applications of the Taylor pattern of Lemma 5.3(i), taken to second order, give the exact identity (odd $n \geq 5$)

$$v^{3/2} = -\frac{5}{64}n^{27/8} + \frac{9}{32}mn^{15/8} - \frac{45}{64}m^2n^{3/8} + \frac{15}{64}vn^{9/8} + \frac{45}{32}vmn^{-3/8} - \frac{9}{64}vm^2n^{-15/8} + \text{err},$$

$|\text{err}| \leq \frac{3}{4}n^{-9/8}$: the fourth-letter phase is a polynomial of degree $(2, 1)$ in the integer pair (m, v) with smooth coefficients. Its v -linear part is vW with W in the family of Theorem 5.11; the vm - and vm^2 -cross terms reduce to the same family by $\xi vm = \xi X v - \xi \theta v$ and $\xi_2 vm^2 = \xi_2 X^2 v - 2\xi_2 X \theta v + \xi_2 \theta^2 v$, whose θ -cross coefficients are sub-unit and whose smooth v -coefficients $\xi X, \xi_2 X^2 \asymp kP^{9/8}$ are family members. Writing $v = Y - \theta_2$ splits each family term into a smooth-in- m part and a kernel factor $e(-c\theta_2)$. The double differencing of Theorem 5.11 applied to the whole mode phase carries the passengers along: the pure- m polynomial phases difference into the Step-3 classes; the $\frac{i}{2}X$ - and $\frac{j}{2}Y$ -passengers difference to curvatures $\ll ih_1 h_2 P^{-5/2}$ and $\asymp jP^{-5/4}$, subdominant in the established hierarchy; the kernel factor is handled by Steps 2–5 verbatim. Assembly and summation over $k \leq J_3$ with $1/k$ -weights give $\sum_k \frac{1}{k} P^{1-1/72} + P^{1-1/24} \ll P^{1-1/72+\varepsilon}$; dyadic blocks sum to $N^{1-1/72+\varepsilon}$. $\square$

The certified-descent density stays $13/16$ at four steps — $OOO*$ is non-contracting at depth 4 ($3^3 > 2^4$) — but the kernel theorem opens the contracting words at depths five and seven.

Theorem 5.14 (the length-5 contracting splits).

$$\#OOOEE(N), \#OOOEO(N) = \frac{N}{32} + O(N^{1-1/72+\varepsilon}), \quad \#OOEOE(N), \#OOEOO(N) = \frac{N}{32} + O(N^{43/48+\varepsilon}).$$

Proof. $OOOE*$. The class indicator is the Theorem-5.13 indicator times $\frac{1}{2}(1 \pm \psi(z^{1/2}))$, $z = J^3(n)$: branch-consistent because on the $OOOE$ cylinder z is even, so the fifth letter is the even-branch value. Vaaler-expand the fifth wave at truncation $P^{1/24}$. The Lemma 5.3(i) pattern along the chain replaces $\frac{1}{2}z^{1/2}$ by $\frac{1}{2}n^{27/16} - \frac{9l}{16}n^{3/16}\theta$ at absorbable decaying cost. The θ -sawtooth has coefficient $\asymp lP^{3/16} < P$; its shifted-window expansion (drift-1 length $P^{13/16}/l$, window $T = P^{1/8} \ll lP^{3/16}$) produces X -modes of size $\asymp lP^{3/16}$ — ordinary first-letter passengers within Theorem 5.13's budget — and the smooth chirp $e(\frac{l}{2}n^{27/16})$ has curvature $\asymp lP^{-5/16}$, subdominant to every retained scale. Theorem 5.13 applies verbatim.

$OOEO*$. On the $OOEO$ cylinder the fifth-letter phase is $\frac{k}{2}w^{3/2}$ with $w = J^3(n)$ odd. Exact linearization writes it as $\frac{k}{2}n^{27/16} - C\theta - B\theta_w$ with $B = \frac{3k}{4}v^{1/4} \asymp kn^{9/16}$, $C = \frac{9k}{16}n^{3/16}$, and $\theta_w = \{v^{1/2}\}$. Expand $e(-B\theta_w)$ on drift-1 intervals of length $P^{7/16}/k$ (there are $\asymp kP^{9/16}$ of them; window $T_w = P^{1/4} \ll kP^{9/16}$, majorant $P^{3/4}$). At frequency $-B + t$, $|t| \leq T_w$, the combined phase is $-\frac{k}{4}v^{3/4} + t v^{1/2}$ plus passengers; linearizing both powers, the θ -coefficients cancel at the window centre up to a residual $\asymp kn^{3/16}$, leaving one slow θ -sawtooth, expanded on the same intervals (window $P^{1/8}$, majorant $P^{7/8}$). The resulting smooth phase has curvature $\lambda_2 = (-\frac{297}{1024} + \frac{27}{128})kn^{-5/16}(1 + o(1)) \asymp kn^{-5/16}$, single-signed since $-0.290k + 0.211k \neq 0$. Van der Corput II on each interval gives $\ll k^{-1/2}P^{9/32}$; over $\asymp kP^{9/16}$ intervals, $S_k \ll k^{1/2}P^{27/32+\varepsilon}$. Balancing $J_5^{1/2}P^{27/32} = P/J_5$ at $J_5 = P^{5/48}$ gives $P^{43/48}$; dyadic blocks sum to $N^{43/48+\varepsilon}$. $\square$

Theorem 5.15 (the length-7 engine splits). All eight classes

$$\#OOEOO**(N), \#OOOEO**(N) = \frac{N}{128} + O(N^{43/48+\varepsilon}).$$

Proof. $OOEOO**$. With $w = J^3(n)$ odd, $p = \lfloor w^{3/2} \rfloor$, $\theta_p = \{w^{3/2}\}$, the exact rearrangement

$$p^{3/2} = -\frac{5}{4}v^{9/8} + \frac{9}{4}wv^{5/8} - \frac{3}{2}w^{3/4}\theta_p + E, \quad 0 \leq E \leq \frac{3}{8}p^{-1/2} + \frac{45}{32}v^{1/8},$$

obtained from two Taylor steps and the identity $\theta_w = v^{1/2} - w$, eliminates the supercritical naive coefficient $\frac{9}{4}v^{5/8} \asymp n^{45/32} > n$ of θ_w ; the remaining sawtooth θ_p has coefficient $\asymp n^{27/32} < n$ and derivative < 1 . Expanding the smooth powers of v by the linearization pattern: the first-letter content is a chirp of curvature $\asymp n^{17/32}$ (van der Corput II); the leading θ_2 -content has coefficient $\asymp n^{33/32}$ — a family of Corollary 5.12 at $\alpha = \frac{33}{32} < \frac{9}{8}$; the remaining θ_2 -amplitudes are $O(n^{9/32})$, inside the engine. The integer block $e(\xi w)$ with $\xi \asymp n^{45/32}$ is handled on frozen gap runs: on the $OOEO$ cylinder $U = v^{1/2}$ has $U'' \asymp P^{-7/8} < 1$, so $\lfloor \Delta U \rfloor$ is constant on runs of length $\asymp P^{7/8}/h$, and per run $\Delta w = J_U + \kappa_w$ with J_U frozen, $e(\xi J_U)$ a smooth chirp, and κ_w an indicator weight — the Lemma 5.10(iii) pattern one level up. The sawtooth θ_p expands on drift-1 intervals of length $P^{5/32}$ with window $T \ll P^{27/32}$, producing X -modes within Theorem 5.14's budget. The seventh letter is the even-branch square root of $q = \lfloor p^{3/2} \rfloor$, with decaying amplitudes only. Theorem 5.14 applies as a passenger theorem.

$OOEOO^{**}$. The analogous rearrangement at base $z^{1/2}$ ($s = \lfloor z^{1/2} \rfloor$: $s^{3/2} = -\frac{1}{2}z^{3/4} + \frac{3}{2}s z^{1/4} + E$) plus the $z^{3/4} \rightarrow n^{81/32}$ linearization chain gives the same four classes of terms — chirp, Corollary-5.12 family at $\alpha = \frac{33}{32}$, engine sawtooths, and an integer- s block on $\lfloor \Delta z^{1/2} \rfloor$ -runs of length $\asymp P^{5/16}$. Same exponent. $\square$

Corollary 5.16 (certified-descent densities 7/8 and 57/64). The class of starts carrying a uniform power-envelope descent certificate of length at most five — evens, OE , $OOEE$, $OOOEE$, $OOEOE$ — has natural density 7/8; adding the length-7 words $OOEOOEE$ and $OOOEOEE$ raises the density of the length- ≤ 7 certified class to 57/64.

Proof. Densities $\frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32} = \frac{7}{8}$ and $\frac{7}{8} + \frac{1}{128} + \frac{1}{128} = \frac{57}{64}$. The cylinders are disjoint, and each new cylinder is counted with a power saving by Theorems 5.14 and 5.15. Contraction is Corollary 2.3: $3^3 = 27 < 32 = 2^5$ on the length-5 words and $3^4 = 81 < 128 = 2^7$ on the length-7 words. $\square$

The leftover $\frac{7}{64}$ is the $OOOO^*$ tree (uncounted) together with the expanding siblings of the counted words. Every uncounted contracting word now passes through $OOOO^*$, whose fifth letter carries the level-3 kernel isolated in Section 6; the first such contractor is $OOOOEEE$.

As with Corollary 5.2, these are densities of uniform certificate classes. They are not densities of all descent certificates and not densities of starts that reach 1.

6. The remaining gap

Theorem 4.1 and Corollary 5.2 say that a uniform one- or two-step argument covers a set of density 3/4; Corollaries 5.9 and 5.16 raise the uniformly certified class to 13/16 at four steps, 7/8 at five, and 57/64 at seven. The first odd-to-odd image expands, so ordinary strong induction cannot fire on the complement. Proposition 4.4 shows that most odd-to-odd starts in a finite window still return below the start inside twenty steps. That is an observation, not Terras's theorem for J .

The depth-by-depth counting of Section 5 does assemble into a conditional Terras-style statement, and the reduction is unconditional:

Proposition 6.1 (equidistribution implies density-one descent). Let $d \geq 1$ and suppose that for every itinerary word w of length d (over all starts, first letter the parity of n),

$$|\#\{n \leq N : \text{word}_d(n) = w\} - 2^{-d}N| \leq E_d(N).$$

Then the starts with no contracting prefix of length $\leq d$ number at most

$$e^{-cd}N + 2^d E_d(N), \quad c = 2\left(\frac{\log 2}{\log 3} - \frac{1}{2}\right)^2 > 0.0342.$$

Every other start $n \geq 2$ satisfies $J^t(n) < n$ for some $t \leq d$ with the uniform power-envelope certificate of Corollary 2.3. Consequently, if $E_d(N) = O_d(N^{1-\delta_d})$ with $\delta_d > 0$ for every d , the set of starts admitting a finite descent certificate has natural density 1.

Proof. A word w of length d has a contracting prefix iff $3^{o_t} < 2^t$ for some $t \leq d$, where o_t counts odd letters among the first t . If w has no contracting prefix then $3^{o_d} \geq 2^d$, i.e. $o_d \geq \beta d$ with $\beta = \log 2 / \log 3 = 0.6309 \dots$. The number of such words is at most $2^d \Pr[\text{Bin}(d, \frac{1}{2}) \geq \beta d] \leq 2^d e^{-2(\beta-1/2)^2 d}$ by Hoeffding's inequality. Each word class has at most $2^{-d}N + E_d(N)$ members; summing over the non-contracting words gives the count. The density-one statement follows by letting $d \rightarrow \infty$ slowly with N (any $d(N) \rightarrow \infty$ with $2^d E_d(N)/N \rightarrow 0$). $\square$

Every E-rooted word has a contracting prefix at length one ($3^0 < 2$), so the hypothesis only ever consumes O-rooted class bounds. Section 5 now proves it *unconditionally at every depth* $d \leq 4$ (Theorems 5.4–5.13), so the base cases of Proposition 6.1 are theorems. The first open case is depth 5: the $OOOO*$ split. It has an exact shape, one nesting deeper than Theorem 5.11. Write $Z = v^{3/2}$, $z = \lfloor Z \rfloor$, $\theta_3 = Z - z$; after four odd letters the fifth is the parity of $\lfloor z^{3/2} \rfloor$.

Lemma 6.2 (level-3 kernel reformulation). For odd $n \geq 5$,

$$\frac{1}{2}(v^{9/4} - z^{3/2}) - \frac{3}{4}z^{1/2}\theta_3 = R_3, \quad 0 \leq R_3 \leq \frac{3}{16}z^{-1/2}.$$

Consequently the fifth-letter phase is, up to $kR_3 \ll kn^{-27/16}$ per term, the exponential sum of the level-3 local floor defect, with coefficient $c = \frac{3k}{4}z^{1/2} \asymp kn^{27/16}$: Lemma 5.10(i) with (m, v) replaced by (v, z) .

Proof. Taylor of $(z + \theta_3)^{3/2}$ at z , with $Z^{3/2} = v^{9/4}$. $\square$

For smooth c with $c \asymp kP^{27/16}$ and $c' \asymp kP^{11/16}$ on $n \sim P$ (the $z^{1/2}$ -shaped family), define

$$K_3(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n) \{ \lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor^{3/2} \}).$$

Conjecture 6.3 (level-3 kernel cancellation). $K_3(P) \ll P^{1-\delta}$ for some $\delta > 0$, uniformly over the family above with $k \leq P^\epsilon$. Exact-phase probes show square-root cancellation ($|K_3| \approx 30.1, 59.5, 423.7$ on $2.5 \cdot 10^3, 2.5 \cdot 10^4, 10^5$ terms), but no proof is claimed.

The boundary between Theorem 5.11 and Conjecture 6.3 is quantitative and sharper than a derivative count. The smooth model iterates: where Theorem 5.11 used two Weyl differencings against $Y'' \asymp P^{1/4} \gg 1 > P^{-3/4} \asymp Y'''$, the level-3 model $n^{27/8}$ has $G''' \asymp P^{3/8} \gg 1 > P^{-5/8} \asymp G^{(4)}$ and predicts three. But the prediction does not descend to the nested floors. The kernel weight now has $c' \asymp kP^{11/16} \gg 1$, so no drift-1 interval exists for any expansion of the weight itself; the branch decomposition of Lemma 5.10(iii) has no analogue at the v -level, because v jumps by $\asymp n^{5/4}$ per step and the branch set is a *product* of two carry lattices, not a copy of one; and the forced inner linearization $v^{3/2} = m^{9/4} - \frac{3}{2}m^{3/4}\theta_2 + E_2$ trades θ_3 for a sawtooth family at coefficient scale $kn^{45/16}$, *above* the $9/4$ threshold where every method of this paper stops. Three routes — a scale-invariant copy of Theorem 5.11, differencing the increment first, and absorbing the leftover into a freezing integer — fail at exactly characterized points; the failures are recorded in the repository as exact statements, not anecdotes.

What survives at the frontier is a model problem and one theorem about it. Stripping every Juggler-specific structure (carries, defects, nesting) from K_3 leaves the *amplitude-product* sums

$$S = \sum_{t \leq L} e(A(t) \{B(t)\}), \quad 1 \ll A' \ll A,$$

with smooth monomial-type A, B (the Juggler instance has $A \asymp P^{27/16}$, $A' \asymp P^{11/16}$). For $A' \ll 1$ partial summation makes the amplitude a tame passenger and the classical Piatetski–Shapiro machinery applies; for $A' \gg 1$ we know of no nontrivial deterministic bound by any method. The generic statement, however, is a theorem — and its proof uses no harmonic analysis at all:

Theorem 6.4 (shift-averaged square-root cancellation). Let $A_1 < \dots < A_L$ be reals with $|A_t - A_{t'}| \geq A'_{\min}|t - t'|$ for some $A'_{\min} \geq 1$, and let $x_1, \dots, x_L$ be arbitrary reals. For $S_\lambda =$

$$\sum_{t \leq L} e(A_t \{x_t + \lambda\}),$$

$$\left| \int_0^1 |S_\lambda|^2 d\lambda - L \right| \leq \frac{6}{\pi} \frac{L}{A'_{\min}} (\log L + 1).$$

In particular $|S_\lambda| \leq \sqrt{L/\varepsilon}$ outside a shift set of measure $\varepsilon(1 + o(1))$: two-sided square-root cancellation for almost every shift of the fractional argument, regardless of the sequence B .

Proof. Expand the square; the diagonal gives L . For $t \neq t'$, the function $\varphi(\lambda) = A_t \{x_t + \lambda\} - A_{t'} \{x_{t'} + \lambda\}$ is piecewise linear on $[0, 1)$ with *real* slope $A_t - A_{t'}$ on at most three arcs (jumps at $1 - \{x_t\}$ and $1 - \{x_{t'}\}$); on each arc $|\int e(\varphi) d\lambda| \leq 1/(\pi |A_t - A_{t'}|)$. Summing, $\sum_{t \neq t'} 3/(\pi A'_{\min} |t - t'|) \leq (6/\pi)(L/A'_{\min})(\log L + 1)$. $\square$

The mechanism deserves note: the amplitude separation $A' \gg 1$ — the very property that defeats every character expansion, since a Fourier window centred at the amplitude drifts by A' harmonics per step — is exactly what makes the shift average trivial. The theorem also forces the empirical profile: censuses of $R = |S|^2/L$ for the Juggler instance at $P = 10^6$ – 10^{10} show a textbook $\text{Exp}(1)$ distribution, now explained rather than merely observed. What remains of Conjecture 6.3 after Theorem 6.4 is a de-randomization: the deterministic sum is the single shift $\lambda = 0$ in a family that cancels almost surely.

Conjecture 6.5 (the pure amplitude-product model). $|S| \leq L^{1-\delta}$ for some $\delta > 0$, for smooth monomial-type A, B with $1 \ll A' \ll A$ as above.

Three structural facts locate the difficulty of the de-randomization, each a short rigorous argument recorded in the repository. No second averaging variable exists: amplitude separation forces any two sample points with $|A(p) - A(q)| \leq 1$ to coincide, and every family average available in the application re-enters either the differenced-kernel class or the amplitude-product class itself. The inverse theory is self-similar: any concentration or discrepancy inverse for $A\{B\} \bmod 1$ is a statement about $\sum e(jA\{B\})$ — the same class with amplitude jA — so the class is closed under its own inverse theorems and no bootstrapping is possible. And the metric statement does not transfer: $|dS_\lambda/d\lambda| \leq 2\pi A_{\max} L$ almost everywhere, so S_λ decorrelates at shift scale $1/A_{\max}$ ($\asymp P^{-27/16}$ in the instance, confirmed by measurement), and an almost-all- λ theorem leaves $\asymp \varepsilon A_{\max}$ exceptional cells among $\asymp A_{\max}$ — no measure argument pins $\lambda = 0$. The deterministic content of Conjecture 6.5 is a specific-point-in-metric-theory problem, the same species as the normality of $\sqrt{2}$: a class whose known successes all use special arithmetic. The instance does carry arithmetic the model forgets — $A^2 = \frac{9}{16}k^2z$ with z integer, coupled to the argument through $\theta_3 = v^{3/2} - z$ — but its natural exploitations (a pure-phase identity, quadratic-field periodicity) re-enter the harmonic toolkit and fail at the recorded points.

A mixed-parity heuristic, ignoring floors, gives mean log-log drift $\frac{1}{2} \log(3/4) < 0$. Finite ensembles sit near this value; hard paths are more odd-rich. This agrees qualitatively with the juggler-like random-walk model of Prasad and Prasad [12]. Fair parity is an assumption, not a dynamical theorem, and typical negative drift is not pointwise contraction.

The gap also resists the obvious finite-state attacks, and the failures are recorded, not anecdotal. Lean certifies a chain of four consecutive expanding persistent residual blocks,

$$1999 \rightarrow 5169 \rightarrow 50093 \rightarrow 193753 \rightarrow 887471$$

(`four_block_pe_1999`), so no uniform run bound of four or less exists for expanding blocks. Around such hard paths, five natural finite-state reductions fail on exact-integer counterexamples: a uniform

slack tax per expanding block fails at the *OOE* block near $1.80 \cdot 10^{32}$ with relative slack below 10^{-30} (Section 2); a compact state built from the normalized landing position θ fails because θ fills essentially all of $[0, 1]$ on continuations and predicts the next parity no better than a residue class; recognition modulo 2^m fails because every residue class modulo 64 both continues and exits the iterated odd-landing sets; a 2-adic restriction from persistent-expanding history fails because the endpoint 763 has $v_2(y^3 - J(y)^2) = 1$; and the expanding word grammar is not an independent obstruction because, on $n \geq 2$, persistence already implies expansion, so that attack reduces to the tautology $J^{|w|}(n) > n$. Each failed reduction is an exact computation on named witnesses.

The analytic routes around Conjecture 6.3 fail at recorded points as well. Composing gap cells across two levels fails: on a cell where the first-level gap is constant, the second-level gap takes a new value at essentially every point (distinct-value ratio 1.0000 in exact windows), so no usable sub-cell survives. Reindexing by the image m (each m has at most one odd preimage) strips one nesting level but introduces a fiber indicator of density $\asymp m^{-1/3}$, whose sawtooth produces mode sums that must beat the sparsity exponent $1/3$ while the engine saves only $1/24$; already the first mode is fatal. At level 3, every character expansion of the kernel product has its Fourier window centred at the amplitude $kc(t)$, which drifts by $kc' \asymp kP^{11/16}$ harmonics per step, so inner sums at fixed harmonic are shorter than one step at every block length; and every algebraic re-form — floor-splitting, the pure-phase identity, differencing at any order, interval-splitting — transfers the $P^{27/16}$ amplitude instead of destroying it. All routes are parked as negative knowledge, not retried.

The gap is therefore:

No theorem forces every exact integer state into a contracting prefix. In particular, it is open whether almost every odd-to-odd start has a finite descent certificate. By Proposition 6.1 that would follow from all-depth parity equidistribution, which is now a theorem through depth four; the first open case is the *OOOO** kernel of Conjecture 6.3, whose model form cancels for almost every shift (Theorem 6.4) and whose deterministic instance is Conjecture 6.5.

That is the Juggler form of the Terras question. A sharper ambient discrepancy exponent does not answer it; by Proposition 6.1 the question now rests on equidistribution at growing depth, beginning with Conjecture 6.3.

7. Software archive

Lean proofs of the exact-arithmetic theorems live at https://github.com/sneakyweasel/balanced_ternary/. The archive is not required to read the arguments above. The analytic estimates of Sections 5 and 6 are human proofs; Lean certifies only the exact floor reductions beneath them (`GapCells.lean`, including the double-gap identity `seq_floor_gap_second` used by Theorem 5.11) and the exact-arithmetic theorems of Sections 2–4. The scaled-integer validators for the linearization lemmas, the kernel probes, and the shift-average and pure-model censuses live in `src/research/juggler_sequence/two_step_parity.py` with pinned tests.

From a clone, the review object is

```
cd formal && lake build Problems.JugglerPaper
```

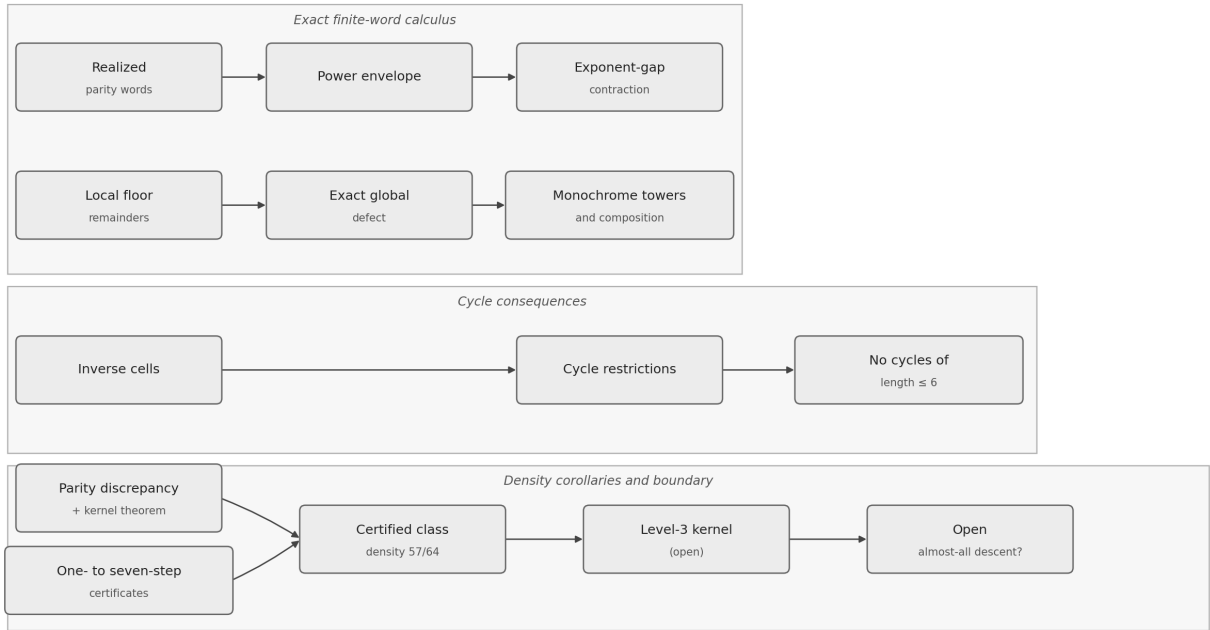



Figure 1: The theorem flow of the note. Exact finite-word identities yield contraction, rigidity, and cycle restrictions; the discrepancy calculus with the kernel theorem counts every itinerary class through depth four and the contracting words of lengths five and seven (certified density $57/64$), leaving the level-3 kernel — generically cancelling by the shift-average theorem — and with it almost-all descent, open.

Acknowledgments

I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. Lean theorems and named computations are the certificates for those claims. The discrepancy estimates of Sections 5 and 6, including the kernel theorem, and Proposition 6.1 are human proofs using classical analytic inequalities; they are not Lean-certified. I take full responsibility for the contents.

References

1. C. A. Pickover, *Computers and the Imagination: Visual Adventures Beyond the Edge*, St. Martin's Press, New York, 1991, ch. 40, p. 232.
2. OEIS Foundation Inc., “Juggler sequence: if n even then $\lfloor \sqrt{n} \rfloor$ else $\lfloor n^{3/2} \rfloor$,” Sequence A094683 in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A094683> (accessed 28 August 2026).
3. J. C. Lagarias, “The $3x+1$ problem and its generalizations,” *Amer. Math. Monthly* 92 (1985), 3–23. doi:10.1080/00029890.1985.11971528.
4. J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x+1$ Problem*, American Mathematical Society, Providence, RI, 2010.
5. R. Terras, “A stopping time problem on the positive integers,” *Acta Arith.* 30 (1976), 241–252. doi:10.4064/aa-30-3-241-252.
6. C. J. Everett, “Iteration of the number-theoretic function $f(2n) = n$, $f(2n+1) = 3n+2$,” *Adv. Math.* 25 (1977), 42–45. doi:10.1016/0001-8708(77)90087-1.
7. T. Tao, “Almost all orbits of the Collatz map attain almost bounded values,” *Forum Math. Pi* 10 (2022), e12. doi:10.1017/fmp.2022.8.
8. R. E. Crandall, “On the “ $3x+1$ ” problem,” *Math. Comp.* 32 (1978), 1281–1292. doi:10.1090/S0025-5718-1978-0480321-3.
9. K. R. Matthews and A. M. Watts, “A generalization of Hasse’s generalization of the Syracuse algorithm,” *Acta Arith.* 43 (1984), 167–175. doi:10.4064/aa-43-2-167-175.
10. L. Kuipers and H. Niederreiter, *Uniform Distribution of Sequences*, Wiley-Interscience, New York, 1974.
11. H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications 53, Providence, RI, 2004.
12. V. Prasad and M. A. Prasad, “Estimates of the maximum excursion constant and stopping constant of juggler-like sequences,” ResearchGate preprint, 2025. doi:10.13140/RG.2.2.14110.04168.
13. J. D. Vaaler, “Some extremal functions in Fourier analysis,” *Bull. Amer. Math. Soc. (N.S.)* 12 (1985), 183–216. doi:10.1090/S0273-0979-1985-15349-2.
14. S. W. Graham and G. Kolesnik, *Van der Corput’s Method of Exponential Sums*, London Mathematical Society Lecture Note Series 126, Cambridge University Press, Cambridge, 1991.